are finitely axiomatizable, and all other finite magmas satisfying these laws are isomorphic to powers of $\mathcal{M}$.

5.2. **Linear models.** As it turns out, a particularly fruitful source of counterexamples is the class of *linear magmas*, where the carrier M is a ring (which may be commutative or noncommutative, finite or infinite), and the operation $\diamond$ is given by $x \diamond y = ax + by$ for some coefficients $a, b \in M$; one can also generalize this slightly to *affine magmas*, in which the operation is given by $x \diamond y = ax + by + c$, but for simplicity we shall focus on linear magmas here. It is easy to see that in a linear magma, any word $w(x_1, \dots, x_n)$ of n indeterminates also takes the linear form

$$w(x_1, \dots, x_n) = \sum_{i=1}^{n} P_{w,i}(a, b) x_i$$

for some (possibly noncommutative) polynomial $P_{w,i}$ in a, b with integer coefficients. Thus, a linear magma will satisfy an equational law $w_1 \simeq w_2$ if and only if the pair (a, b) lies in the (possibly noncommutative) variety

$$V_{w_1,w_2}(M) \coloneqq \{(a, b) \in M \times M : P_{w_1,i}(a, b) = P_{w_2,i}(a, b) \text{ for all } i\} \subseteq M^2. \tag{3}$$

As such, a necessary condition for such a law $w_1 \simeq w_2$ to entail another law $w_1' \simeq w_2'$ is that one has the inclusion

$$V_{w_1,w_2}(M) \subseteq V_{w_1',w_2'}(M)$$

for all rings M. For commutative rings, this criterion can be checked in an automatable fashion by standard Gröbner basis techniques; in the noncommutative case one can use methods such as the diamond lemma [10].

Example 5.2 (Commutative counterexample)**.** For the law $\mathrm{x} \simeq \mathrm{y} \diamond (((\mathrm{x} \diamond \mathrm{y}) \diamond \mathrm{x}) \diamond \mathrm{y})$ (E1286), the variety (3) can be computed to be

$$\{(a, b) \in M \times M : 1 = ba^3 + bab, 0 = a + ba^2b + b^2\}$$

while the variety for the idempotent law E3 is

$$\{(a, b) \in M \times M : a + b = 1\}.$$

Thus, to show that E1286 does not entail E3, it suffices to locate elements a, b of a ring M for which one has $1 = ba^3 + bab$, $0 = a + ba^2b + b^2$, and $a + b \neq 1$. Here one can take a commutative example, for instance when $M = \mathbb{Z}/p\mathbb{Z}$ and $(p, a, b) = (11, 1, 7)$.

Example 5.3 (Noncommutative counterexample)**.** For the law $\mathrm{x} \simeq \mathrm{y} \diamond ((\mathrm{y} \diamond (\mathrm{x} \diamond \mathrm{z})) \diamond \mathrm{z})$ (E1117), the variety (3) can be computed to be

$$\{(a, b) \in M \times M : 1 = baba, 0 = a + ba^2, 0 = bab^2 + b^2\}$$

while the variety for $\mathrm{x} \simeq (\mathrm{x} \diamond ((\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{x})) \diamond \mathrm{x}$ (E2441) is

$$\{(a, b) \in M \times M : 1 = a^2 + aba^2 + abab + ab^2 + b\}.$$

Observe that if $ba = -1$, then (a, b) automatically lies in the first set, and lies in the second set if and only if $(ab+1)(b-1) = 0$. One can then show that E1117 does not imply E2441 by

is sufficient to show that all other finite magmas satisfying the laws of $\mathcal{M}$ are isomorphic to products of submagmas of $\mathcal{M}$. By combining these observations with [58], one can also show that quasiprimal finite magmas can be axiomatized by a single equation. We thank Stanley Burris for these comments, as well as the other references and observations in this remark.

setting $a = L$, $b = -R$ where L, R are the left and right shift operators respectively on the ring of integer-valued sequences $\mathbb{Z}^{\mathbb{N}}$. With some *ad hoc* effort one can convert this example into a less linear, but simpler (and easier to formalize) example, namely the magma with carrier $\mathbb{Z}$ and operation $x \diamond y = 2x - \lfloor y/2 \rfloor$.

Remark 5.4. As essentially observed in [5], if there is a commutative linear counterexample to an implication $\mathrm{E} \models \mathrm{E}'$, then by the Lefschetz principle this counterexample can be realized in a finite field $\mathbb{F}_q$ for some prime power q (and by the Chebotarev density theorem one can in fact take q to be a prime, so that the carrier is of the form $\mathbb{Z}/p\mathbb{Z}$ for some prime p), so that one also has $\mathrm{E} \models_{\mathrm{fin}} \mathrm{E}'$. As such, we have found that an effective way to refute implications by the commutative linear magma method is to simply perform a brute force search over linear magmas $x \diamond y = ax + by$ in $\mathbb{Z}/p\mathbb{Z}$ for various triples (p, a, b).

On the other hand, the refutations obtained by noncommutative linear constructions need not have a finite model. For instance, consider the refutation E1117 $\not\models$ E2441 from Example 5.3. The law E1117 can be rewritten as $L_y R_z L_y R_z x = x$. This implies that R_z is injective and L_y is surjective for all y, z. For finite magmas $\mathcal{M}$, this then implies that the L_y, R_z are in fact invertible, and hence we have also $R_z L_y R_z L_y x = x$, which implies E2441 by setting $x = y = z$. Thus, the refutation E1117 $\not\models$ E2441 is "immune" to finite counterexamples.

Remark 5.5. One can also consider nonlinear magma models, such as quadratic models $x \diamond y = ax^2 + bxy + cy^2 + dx + ey + f$ in a cyclic group $\mathbb{Z}/N\mathbb{Z}$. For small values of N, we have found such models somewhat useful in providing additional refutations of implications $\mathrm{E} \models_{\mathrm{fin}} \mathrm{E}'$ beyond what can be achieved by the linear or affine models. However, as the polynomials associated to a word $w(x_1, \dots, x_n)$ tend to be of high degree (exponential in the order of the word), it becomes quite rare for such models to satisfy a given equation E when N is large.

Remark 5.6. One can also consider the seemingly more general linear model $x \diamond y = ax + by$, where the carrier M is now an abelian group, and a, b act on M by homomorphisms, that is to say that they are elements of the endomorphism ring $\mathrm{End}(M)$. However, this leads to exactly the same varieties (3) (where M is now replaced by the endomorphism ring $\mathrm{End}(M)$) and so does not increase the power of the linear model for the purposes of refuting implications.

On the other hand, there are certainly some refutations $\mathrm{E} \not\models \mathrm{E}'$ of implications that are "immune" to both commutative and noncommutative linear models, in the sense that all such models that satisfy E, also satisfy E′. One such example is the refutation E1485 $\not\models$ E151, which we discuss further in Section 5.4 below.

5.3. **Translation-invariant models.** It is natural to look for counterexamples amongst magmas that satisfy a large number of symmetries. One such class of counterexamples are *translation-invariant models*, in which the carrier M is a group, and the left translations of this group form isomorphisms of the magma $\mathcal{M}$. In the case of an abelian group $M = (M, +)$, such models take the form

$$x \diamond y = x + f(y - x) \tag{4}$$

for some function $f \colon M \to M$; in the case of a non-abelian group $M = (M, \cdot)$, such models instead take the form

$$x \diamond y = x f(x^{-1} y).$$

For such models, the verification of an equational law in n variables corresponds to a functional equation for f in $n-1$ variables, as the translation symmetry allows one to normalize one variable to be the identity (say). This can simplify an implication to the point where an explicit counterexample can be found. These functional equations are trivial to analyze when $n = 1$. For $n = 2$, they are not as trivial, but still quite tractable, and has led to several refutations in practice. The method does not appear to be particularly effective for $n > 2$ due to the complexity of the functional equations.

Example 5.7 (Abelian example)**.** For the law $\mathrm{x} \simeq (\mathrm{x} \diamond \mathrm{y}) \diamond ((\mathrm{x} \diamond \mathrm{y}) \diamond \mathrm{y})$ (E1648), we apply the abelian translation-invariant model (4) with $y = x + h$ to obtain

$$\begin{aligned} x \diamond y &= x + f(h) \\ (x \diamond y) \diamond y &= x + f(h) + f(h - f(h)) \\ (x \diamond y) \diamond ((x \diamond y) \diamond y) &= x + f(h) + f(f(h - f(h))) \end{aligned}$$

so that this model satisfies E1648 if and only if the functional equation

$$f(h) + f(f(h - f(h))) = 0$$

holds for all $h \in M$. Similarly, the law $\mathrm{x} \simeq (\mathrm{x} \diamond (\mathrm{x} \diamond \mathrm{y})) \diamond \mathrm{y}$ (E206) is satisfied if and only if

$$f(f(h)) + f(h - f(f(h))) = 0$$

for all $h \in M$. One can now check that the function $f \colon \mathbb{Z} \to \mathbb{Z}$ defined by $f(h) := -\mathrm{sgn}(h)$ (thus $f(h)$ equals -1 when h is positive, $+1$ when h is negative, and 0 when h is zero) satisfies the first functional equation but not the second, thus establishing that E1648 $\not\models$ E206.

Example 5.8 (Non-abelian example)**.** We now obtain the opposite refutation E206 $\not\models$ E1648 to Example 5.7 using the non-abelian translation-invariant model. By similar calculations to before, we now seek to find a function $f \colon M \to M$ on a non-abelian group $(M, \cdot)$ that satisfies the functional equation

$$f(f(h))f(f(f(h))^{-1}h) = 1 \tag{5}$$

for all $h \in M$, but fails to satisfy the functional equation

$$f(h)f(f(f(h)^{-1}h)) = 1 \tag{6}$$

for at least one $h \in M$. Now take M to be the group generated by three generators a, b, c subject to the relations $a^2 = b^2 = c^2 = 1$, or equivalently the group of reduced words in a, b, c with no adjacent letters in the word equal. We define

$$f(1) = 1, f(a) = b, f(b) = c, f(c) = a$$

and then $f(aw) = a$ for any non-empty reduced word w not starting with a, and similarly for b and c. The equation (5) can be checked directly for $h = 1, a, b, c$. If $h = aw$ with w non-empty, reduced, and not starting with a, then $f(f(h))^{-1} = f(f(h)) = b$ and $f(f(f(h))^{-1}h) = f(baw) = b$, giving (5) in this case, and similarly for cyclic permutations. Meanwhile, (6) can be checked to fail for $h = a$.

Remark 5.9. The construction in Example 5.8 also has the following more "geometric" interpretation. The carrier M can be viewed as the infinite 3-regular tree, in which every vertex imposes a cyclic ordering on its 3 neighbors (for instance, if we embed M as a planar graph, we can use the clockwise ordering). For $x, y \in M$, we then define $x \diamond y$ to equal x if $x = y$. If y is instead a neighbor of x, we define $x \diamond y$ to be the next neighbor of x in the cyclic